2–8 weeks to months for antibodies to appear after infection, during part of this period, the individual may be highly infectious. This sero negative infective stage is known as the 'window period'. Antibody can be detected by

1. ELISA
2. Western blot test.

Treatment

The treatment of AIDS include:

1. The treatment and prophylaxis of infections and tumours.
2. General management
3. Immunorestrorative measure
4. Specific anti-HIV agents.

Effective drugs are available, they are Zidovudine, Didanosine, Zalcitabine, Lamivudine and Protease inhibitors like Saquinavir, Ritonavir, Indinavir used as monotherapy or in various combination.

10.7 Arbo Virus

Arbo Viruses (arthropod - borne viruses) are viruses of vertebrates biologically transmitted by **hematophagous insect vectors**. They multiply in blood sucking insects and are transmitted by bite to vertebrate hosts. Arbo viruses are worldwide in distribution. Arbo viruses have been named according to the disease caused (**yellow fever**), the place of isolation of the virus (**kyasanur forest disease**) or the local name for the disease (**chikungunya**). They are classified into Toga, Flavi, Bunya, Reo and Rhabdovirus families. Arbo viruses have a very wide host range including many species of animals and birds. The most important arbo virus vectors are **mosquitoes**, followed by **ticks**.

The virus enters the body through the **bite** of the insect vector. After multiplication in the **reticuloendothelial system, viremia** of varying duration occurs, or the virus is transported to the target **organs** such as central nervous system in **encephalitides**, the liver in **yellow fever** and the capillary endothelium in **hermorrhagic fever**.

Clinical syndromes are fever with or without rash encephalitis, hemorrhagic fever with or without rash encephalitis, hemorrhagic fever and systemic disease, yellow fever.

Diagnosis may be established by virus isolation or serology.

Samples (Blood, CSF) are inoculated intra cerebrally into sucking mice. The animal develop **fatal encephalitis**. Viruses may be isolated in tissue cultures or in eggs. Isolates are identified by hemagglutination inhibition, complement fixation, gel precipitation, immunofluorescence.

ELISA

Virus isolated from insect vectors and from reservoir animal.

Toga Viruses

Toga viruses are **spherical enveloped viruses** with a diameter of 50-70nm. Single stranded RNA genome. The virus replicates in the **cyloplasm** of the host cell and released by budding through host cell membranes. The name Toga Virus is derived from **'toga'** meaning the Roman Mantle refers to the **viral envelope**.

The genus Alpha Virus was formerly classified as **Group A arbo viruses** which explains the name Alpha Virus. The genus Alpha Virus contains 32 species of which 13 infect humans. All are **mosquito borne**.

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